

2024, Mar. 2024 Release LOC:Acute Adult Stroke

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Select Day

Initial review (1, 2, 3, 4, 5)

Episode Day 1 (1, 2, 4, 5)

Episode Day 2 (5, 47, 48)

Episode Day 3 (5, 71, 72)

Episode Day 4 (5, 72)

Episode Day 5

Discharge Screens (78)

Utilization Benchmarks

Notes

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Overview

Instruction:

This subset is for patients with a diagnosis of a stroke (including resolved or resolving minor/nondisabling stroke). For patients with transient ischemic attack see the Transient ischemic attack (TIA) subset. Patients requiring surgical management of an intracranial bleed (e.g., epidural hematoma, subdural hematoma, or subarachnoid hemorrhage), should be reviewed in the General Surgical or General Trauma subset.

Introduction:

Stroke includes any acute focal injury of the central nervous system (CNS) originating from a vascular event. When using the term stroke, the following definitions should be taken into consideration:

- CNS infarction: brain, spinal cord, or retinal cell death caused by ischemia
- Ischemic stroke: episode of neurological dysfunction caused by focal cerebral, retinal, or spinal infarction
- Silent CNS infarction: imaging evidence of CNS infarction without neurological dysfunction
- Imaging evidence for intracranial hemorrhage
- Intracerebral hemorrhage (ICH): collection of blood within the brain parenchyma or ventricular system not caused by trauma
- Minor non-disabling ischemic stroke: characterized by neurological symptoms resolving or resolved
- Stroke caused by ICH: rapidly developing clinical signs of neurological dysfunction related to the collection of blood
- Silent cerebral hemorrhage: imaging evidence of a collection of chronic blood products within the brain parenchyma, subarachnoid space, or ventricular system with no evidence of neurologic dysfunction
- Subarachnoid hemorrhage (SAH): bleeding into the subarachnoid space
- Stroke caused by SAH: rapidly developing clinical signs of neurological dysfunction related to bleeding into the subarachnoid space
- Stroke caused by cerebral venous thrombosis: infarction or hemorrhage into the brain, spinal cord, or retina as a result of thrombosis of a cerebral vessel
- Stroke, not otherwise specified: an acute episode of neurological dysfunction, with the suspected cause of ischemia or intracerebral hemorrhage, unable to be classified as any one of the above

Evaluation and Treatment:

Initial evaluation involves history and physical, electrocardiogram (ECG), and laboratory assessment. Imaging studies

are done to rule out alternative diagnoses and to search for potential sources such as embolism and cerebrovascular disease. The initial workup will vary by the specific disorder and may include:

- Brain imaging: computer tomography (CT), magnetic resonance imaging (MRI), or diffusion-weighted MRI (DW-MRI)
- Vascular imaging: computer tomography angiography (CTA), magnetic resonance angiography (MRA), or carotid ultrasonography
- Cardiac imaging: transthoracic echocardiogram (TTE), transesophageal echocardiogram (TEE), or multidetector cardiac computed tomography

The use of thrombolytics is determined by the duration of symptoms and is contraindicated in the presence of hemorrhage. For all eligible acute ischemic stroke patients, intravenous recombinant tissue-type plasminogen activator (r-tPA) should be given within 4.5 hours of symptom onset. Endovascular therapy with a stent retriever should be considered for candidates with large vessel occlusion stroke even if the patient has received r-tPA. The benefits of both r-tPA and mechanical thrombectomy are time dependent. Ideally, brain imaging should be performed within 20 minutes after a patient with a suspected stroke arrives to the hospital and r-tPA should be started as quickly as possible. In selected patients, mechanical thrombectomy can be beneficial up to 24 hours after symptom onset (Kleindorfer et al., Stroke 2021, 52: e364-e467; Powers et al., Stroke 2019, 50: e344-e418). Hypotension and hypovolemia should be corrected to maintain systemic perfusion levels necessary to support organ function (Powers et al., Stroke 2019, 50: e344-e418). Following an ischemic stroke, antiplatelet administration is recommended within 24 to 48 hours after symptom onset. For patients treated with r-tPA, aspirin is delayed for 24 hours. In patients with a new functional, or worsening of existing impairment on admission, a rehabilitation therapy evaluation should be completed to determine goals and needs for safe discharge planning (National Institute for Health and Care Excellence (NICE). Stroke and transient ischaemic attack in over 16s: diagnosis and initial management. NG128. London: NICE; 2019; Powers et al., Stroke 2019, 50: e344-e418; Powers et al., Stroke 2018, 49: e46-e110).

Many stroke patients are at high risk for readmission and careful attention to discharge planning is essential.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Kleindorfer et al., Stroke 2021, 52: e364-e467; Powers et al., Stroke 2019, 50: e344-e418).

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3, 4, 5)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽⁶⁾

- Neurological deficit resolved or resolving ^(7, 8)
- CT or MRI performed ⁽⁹⁾
- Vascular imaging of carotid artery performed within 24h ⁽¹⁰⁾

● **ACUTE, One:** ⁽¹¹⁾● Suspected acute stroke, **Both:** ^(12, 13, 14)● New and persistent neurological deficit, ≥ **One:** ⁽¹⁵⁾

- Abnormal movement pattern or alteration in muscle tone ⁽¹⁶⁾
- Alteration in speech ⁽¹⁷⁾
- Gait disturbance
- Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽¹⁸⁾
- Paresis or paralysis ⁽¹⁹⁾
- Sensory deficit ⁽²⁰⁾
- Visual impairment ⁽²¹⁾
- Vertigo or dizziness at rest ^(22, 23)

- CT or MRI performed ^(24, 25, 26)

● **CRITICAL, ≥ One:** ⁽²⁷⁾

- Thrombolytic ^(28, 29, 30, 31, 32)
- Endovascular intervention ^(33, 34)
- Cerebral edema

● Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One:** ⁽³⁵⁾

- MAP < 65 mmHg
- Systolic blood pressure < 90 mmHg and < baseline
- Hemodynamic monitoring, invasive ⁽³⁶⁾
- Antihypertensive, continuous ⁽³⁷⁾
- Mechanical ventilation

● NIPPV and, ≥ **One:** ⁽³⁸⁾

- Arterial $\text{Po}_2 \leq 39 \text{ mmHg (5.2 kPa)}$
- Arterial or venous $\text{Pco}_2 \geq 60 \text{ mmHg (8.0 kPa)}$ and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD

- Seizure (excludes epilepsy), requiring anticonvulsant

- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 1, One:** ^(1, 2, 4, 5)

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽⁶⁾

- Neurological deficit resolved or resolving ^(7, 8)
- CT or MRI performed ⁽⁹⁾
- Vascular imaging of carotid artery performed within 24h ⁽¹⁰⁾
- Antiplatelet agent or anticoagulant (includes PO) ^(39, 40)

● **ACUTE, One:** ⁽¹¹⁾● Suspected acute stroke, **All:** ^(12, 13, 14)● New and persistent neurological deficit, ≥ **One:** ⁽¹⁵⁾

- Abnormal movement pattern or alteration in muscle tone ⁽¹⁶⁾
- Alteration in speech ⁽¹⁷⁾
- Gait disturbance
- Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽¹⁸⁾

- Paresis or paralysis ⁽¹⁹⁾
- Sensory deficit ⁽²⁰⁾
- Visual impairment ⁽²¹⁾
- Vertigo or dizziness at rest ^(22, 23)
- CT or MRI performed ^(24, 25, 26)
- Antiplatelet agent or anticoagulant (includes PO), administered or contraindicated ^(40, 41, 42, 43)
- **CRITICAL, ≥ One:** ⁽²⁷⁾
 - Thrombolytic ^(28, 29, 30, 31, 32, 44)
 - Endovascular intervention ^(33, 34)
 - **Cerebral edema and, ≥ One:** ⁽⁴⁵⁾
 - ICP monitoring (includes OR placement)
 - Mannitol or hypertonic saline ≥ 3%(0.03) solution ⁽⁴⁶⁾
 - **Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ One:** ⁽³⁵⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Hemodynamic monitoring, invasive ⁽³⁶⁾
 - Antihypertensive, continuous ⁽³⁷⁾
 - Mechanical ventilation
 - **NIPPV and, ≥ One:** ⁽³⁸⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - **Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
 - Seizure (excludes epilepsy), requiring anticonvulsant
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2, One:** ^(5, 47, 48)
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **Both:** ⁽⁴⁹⁾
 - Neurological stability ⁽⁵⁰⁾
 - Systolic blood pressure within acceptable limits ^(51, 52)
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
 - **Hypertension, Both:**
 - Systolic blood pressure > 180 mmHg
 - Antihypertensive (includes PO) ⁽⁵³⁾
 - **ACUTE, One:**
 - **Early responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁴⁹⁾
 - Afebrile
 - Neurological stability ⁽⁵⁰⁾
 - Systolic blood pressure within acceptable limits ^(51, 52)
 - No complication or active comorbidity relevant to this episode of care ⁽⁵⁴⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - **Stroke, One:**
 - Hemorrhagic stroke and neurological assessment at least 3x/24h ^(55, 56)
 - **Acute ischemic stroke, actual or suspected and, Both:** ^(14, 55, 56)
 - Antiplatelet agent or anticoagulant (includes PO) ^(40, 41, 42, 43)
 - Neurological assessment at least 3x/24h
 - **Bleeding, actual or suspected and, Both:** ⁽⁵⁷⁾
 - **Finding, ≥ One:**
 - Heart rate 100-120/min, sustained ⁽⁵⁸⁾
 - Hct or Hb decreased within last 24h
 - **Orthostatic hypotension, Both:** ⁽⁵⁹⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing

- Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
- **Intervention, $\geq$ One:**
 - Blood product transfusion
 - Hct or Hb monitoring at least 2x/24h, ≤ 2 d
- **Feeding tube, Both:**
 - Insertion performed within last 24h
 - Feedings initiated
- Functional impairment (new) and rehabilitation initiation ≤ 24 h ^(60, 61, 62)
- **Hypertension ≤ 2 d, Both:** ⁽⁵²⁾
 - Systolic blood pressure > 180 mmHg
 - Antihypertensive medication (includes PO) ⁽⁵³⁾
- **Pneumonia confirmed by imaging and, Both:** ⁽⁶³⁾
 - **Finding, $\geq$ One:**
 - Temperature $\geq 100.4^{\circ}\text{F}(38.0^{\circ}\text{C})$ ⁽⁶⁴⁾
 - O_2 sat $\leq 91\%(0.91)$ and $<$ baseline ⁽⁶⁵⁾
 - WBC $\geq 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$
 - WBC $\leq 4,000/\text{cu.mm}(4 \times 10^9/\text{L})$
 - Bands $> 10\%(0.10)$
 - **Intervention, $\geq$ One:** ⁽⁶⁶⁾
 - Requiring supplemental oxygen ⁽⁶⁷⁾
 - Anti-infective
 - Respiratory interventions at least 6x/24h ⁽⁶⁸⁾
- Post Critical care ≤ 24 h
- **INTERMEDIATE, $\geq$ One:**
 - Neurological assessment at least every 3-4h, ≤ 2 d ^(55, 56)
 - **Seizure (excludes epilepsy), activity resolved, One:**
 - Anticonvulsant, continuous ≤ 2 d
 - Anticonvulsant, conversion from IV to PO ≤ 24 h
 - NIPPV ⁽³⁸⁾
 - Oxygen $\geq 40\%(0.40)$, ≤ 2 d ^(67, 69)
- **CRITICAL, $\geq$ One:**
 - Neurological assessment every 1-2h, ≤ 2 d ^(55, 56)
 - **Cerebral edema and, $\geq$ One:** ⁽⁴⁵⁾
 - ICP monitoring (includes OR placement)
 - Mannitol or hypertonic saline $\geq 3\%(0.03)$ solution ⁽⁴⁶⁾
 - Post thrombolytic therapy ≤ 24 h ⁽³⁰⁾
 - Antihypertensive, continuous ⁽³⁷⁾
 - **Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, $\geq$ One:** ⁽³⁵⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Hemodynamic monitoring, invasive ⁽³⁶⁾
 - Mechanical ventilation
 - **NIPPV and, $\geq$ One:** ⁽³⁸⁾
 - $\text{FiO}_2 > 50\%(0.50)$
 - Respiratory intervention every 1-2h ⁽⁶⁸⁾
 - Seizure (excludes epilepsy), recurrent or refractory requiring anticonvulsant ⁽⁷⁰⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 3, One:** ^(5, 71, 72)
 - **OBSERVATION, One:**
 - **Responder, discharge expected today if clinically stable last 12h, Both:** ⁽⁴⁹⁾
 - Neurological stability ⁽⁵⁰⁾
 - Systolic blood pressure within acceptable limits ^(51, 52)

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One**:
 - For a condition other than stroke use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽⁴⁹⁾
 - Afebrile
 - Neurological stability ⁽⁵⁰⁾
 - Systolic blood pressure within acceptable limits ^(51, 52, 73)
 - **Complication or comorbidity, One**:
 - No complication or active comorbidity relevant to this episode of care ⁽⁵⁴⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One**: ⁽⁵⁴⁾
 - Bleeding controlled or resolved
 - **Functional impairment and, One**: ⁽⁶⁰⁾
 - **Discharge to home and, One**: ⁽⁷⁴⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - **Pneumonia and, Both**:
 - O₂ sat within acceptable limits ⁽⁵¹⁾
 - WBC within acceptable limits ⁽⁵¹⁾
 - Tolerating PO or nutritional route established ^(56, 75, 76)
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - **Bleeding, actual or suspected and, Both**: ⁽⁵⁷⁾
 - **Finding, ≥ One**:
 - Heart rate 100-120/min, sustained ⁽⁵⁸⁾
 - Hct or Hb decreased within last 24h
 - **Orthostatic hypotension, Both**: ⁽⁵⁹⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing
 - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
 - **Intervention, ≥ One**:
 - Blood product transfusion
 - Hct or Hb monitoring at least 2x/24h, ≤ 2d
 - **Feeding tube, Both**:
 - Insertion performed within last 24h
 - Feedings initiated
 - Functional impairment (new) and rehabilitation initiation ≤ 24h ^(60, 61, 62)
 - **Hypertension ≤ 2d, Both**: ⁽⁵²⁾
 - Systolic blood pressure > 180 mmHg
 - Antihypertensive (includes PO) ⁽⁵³⁾
 - **Pneumonia confirmed by imaging and, Both**: ⁽⁶³⁾
 - **Finding, ≥ One**:
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁴⁾
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽⁶⁵⁾
 - WBC ≥ 12,000/cu.mm(12x10⁹/L)
 - WBC ≤ 4,000/cu.mm(4 x10⁹/L)
 - Bands > 10%(0.10)
 - **Intervention, ≥ One**: ⁽⁶⁶⁾
 - Requiring supplemental oxygen ⁽⁶⁷⁾
 - Anti-infective
 - Respiratory interventions at least 6x/24h ⁽⁶⁸⁾
 - Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One**:

- Neurological assessment at least every 3-4h, $\leq 2d$ ^(55, 56)
- Seizure (excludes epilepsy), activity resolved, **One:**
 - Anticonvulsant, continuous $\leq 2d$
 - Anticonvulsant, conversion from IV to PO $\leq 24h$
- NIPPV ⁽³⁸⁾
- Oxygen $\geq 40\%(0.40)$, $\leq 2d$ ^(67, 69, 77)
- **CRITICAL, $\geq$ One:**
 - Neurological assessment every 1-2h, $\leq 2d$ ^(55, 56)
 - Cerebral edema and, $\geq$ **One:** ⁽⁴⁵⁾
 - ICP monitoring (includes OR placement)
 - Mannitol or hypertonic saline $\geq 3\%(0.03)$ solution ⁽⁴⁶⁾
 - Antihypertensive, continuous ⁽³⁷⁾
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24h$ and, $\geq$ **One:** ⁽³⁵⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Hemodynamic monitoring, invasive ⁽³⁶⁾
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽³⁸⁾
 - $FiO_2 > 50\%(0.50)$
 - Respiratory intervention every 1-2h ⁽⁶⁸⁾
 - Seizure (excludes epilepsy), recurrent or refractory requiring anticonvulsant ⁽⁷⁰⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 4, One:** ^(5, 72)
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁴⁹⁾
 - Afebrile
 - Neurological stability ⁽⁵⁰⁾
 - Systolic blood pressure within acceptable limits ^(51, 52)
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁵⁴⁾
 - **Complication or comorbidity and clinically stable for discharge, $\geq$ One:** ⁽⁵⁴⁾
 - Bleeding controlled or resolved
 - **Functional impairment and, One:** ⁽⁶⁰⁾
 - **Discharge to home and, One:** ⁽⁷⁴⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - **Pneumonia and, Both:**
 - O_2 sat within acceptable limits ⁽⁵¹⁾
 - WBC within acceptable limits ⁽⁵¹⁾
 - Tolerating PO or nutritional route established ^(56, 75, 76)
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - **Bleeding, actual or suspected and, Both:** ⁽⁵⁷⁾
 - **Finding, $\geq$ One:**
 - Heart rate 100-120/min, sustained ⁽⁵⁸⁾
 - Hct or Hb decreased within last 24h
 - **Orthostatic hypotension, Both:** ⁽⁵⁹⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing
 - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
 - **Intervention, $\geq$ One:**
 - Blood product transfusion
 - Hct or Hb monitoring at least 2x/24h, $\leq 2d$

- **Feeding tube, Both:**
 - Insertion performed within last 24h
 - Feedings initiated
- Functional impairment (new) and rehabilitation initiation ≤ 24h ^(60, 61, 62)
- **Hypertension ≤ 2d, Both:** ⁽⁵²⁾
 - Systolic blood pressure > 180 mmHg
 - Antihypertensive (includes PO) ⁽⁵³⁾
- **Pneumonia confirmed by imaging and, Both:** ⁽⁶³⁾
 - **Finding, ≥ One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁴⁾
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽⁶⁵⁾
 - WBC ≥ 12,000/cu.mm(12x10⁹/L)
 - WBC ≤ 4,000/cu.mm(4 x10⁹/L)
 - Bands > 10%(0.10)
 - **Intervention, ≥ One:** ⁽⁶⁶⁾
 - Requiring supplemental oxygen ⁽⁶⁷⁾
 - Anti-infective
 - Respiratory interventions at least 6x/24h ⁽⁶⁸⁾
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - Neurological assessment at least every 3-4h, ≤ 2d ^(55, 56)
 - **Seizure (excludes epilepsy), activity resolved, One:**
 - Anticonvulsant, continuous ≤ 2d
 - Anticonvulsant, conversion from IV to PO ≤ 24h
 - NIPPV ⁽³⁸⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ^(67, 69, 77)
- **CRITICAL, ≥ One:**
 - Neurological assessment every 1-2h, ≤ 2d ⁽⁵⁵⁾
 - Seizure (excludes epilepsy), recurrent or refractory requiring anticonvulsant ⁽⁷⁰⁾
 - **Cerebral edema and, ≥ One:** ⁽⁴⁵⁾
 - ICP monitoring (includes OR placement)
 - Mannitol or hypertonic saline ≥ 3%(0.03) solution ⁽⁴⁶⁾
 - Antihypertensive, continuous ⁽³⁷⁾
 - **Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ One:** ⁽³⁵⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Hemodynamic monitoring, invasive ⁽³⁶⁾
 - Mechanical ventilation
 - **NIPPV and, ≥ One:** ⁽³⁸⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁶⁸⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 5, One:**
 - **ACUTE, One:**
 - **Responder, discharge expected today if clinically stable last 12h, All:** ⁽⁴⁹⁾
 - Afebrile
 - Neurological stability ⁽⁵⁰⁾
 - Systolic blood pressure within acceptable limits ^(51, 52)
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁵⁴⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One:** ⁽⁵⁴⁾
 - Bleeding controlled or resolved

- Functional impairment and, **One:** ⁽⁶⁰⁾
 - Discharge to home and, **One:** ⁽⁷⁴⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - Pneumonia and, **Both:**
 - O₂ sat within acceptable limits ⁽⁵¹⁾
 - WBC within acceptable limits ⁽⁵¹⁾
 - Tolerating PO or nutritional route established ^(56, 75, 76)
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
 - Use **Extended Stay subset**
 - Refer for secondary review
- **INTERMEDIATE, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
 - Use **Extended Stay subset**
 - Refer for secondary review
- **CRITICAL, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - Use **Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, **One:** ⁽⁷⁸⁾

- **HOME, Both:**
 - Level of care appropriateness, **Both:**
 - Home environment safe and accessible ⁽⁷⁹⁾
 - Patient or caregiver demonstrates ability to manage care
 - Stroke discharge planning, **All:** ^(80, 81)
 - Provide comprehensive written discharge and teaching instructions ⁽⁸²⁾
 - Education on the importance of, **All:**
 - Strict medication adherence ⁽⁸³⁾
 - Follow-up care
 - Smoking cessation and avoiding passive smoke
 - Elimination or reduction of alcohol intake
 - Engaging in at least 30 minutes of physical activity daily
 - Achieving and maintaining a healthy weight
 - Understands when to seek help
 - Medication reconciliation ⁽⁸⁴⁾
 - Follow-up care planned ⁽⁸⁵⁾
 - Identify and address transportation needs
- **HOME CARE, Both:**
 - Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽⁷⁹⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(86, 87)
 - Skilled services, ≥ **One:**
 - Skilled nursing ⁽⁸⁸⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁸⁹⁾
 - Stroke discharge planning, **All:** ^(80, 81)
 - Provide comprehensive written discharge and teaching instructions ⁽⁸²⁾

- Education on the importance of, **All:** ⁽⁵²⁾
 - Strict medication adherence ⁽⁸³⁾
 - Follow-up care
 - Smoking cessation and avoiding passive smoke
 - Elimination or reduction of alcohol intake
 - Engaging in at least 30 minutes of physical activity daily
 - Achieving and maintaining a healthy weight
 - Understands when to seek help
- Medication reconciliation ⁽⁸⁴⁾
- Follow-up care planned ⁽⁸⁵⁾
- Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽⁹⁰⁾
 - Skilled treatment, **≥ One:**
 - Clinical assessment ⁽⁹¹⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁹²⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 1\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment $1\text{-}2\text{x}/24\text{h}$ ⁽⁸⁸⁾
 - Therapy, **All:**
 - Goal-directed therapy with rehabilitation potential ^(93, 94)
 - Functional impairments requiring at least supervision ⁽⁹⁵⁾
 - Able to tolerate 1h/d of skilled therapy $\geq 5\text{d/wk}$
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 2\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment $4\text{h}/24\text{h}$ ⁽⁸⁸⁾
 - Therapy, **All:**
 - Goal-directed therapy with rehabilitation potential ^(93, 94)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹⁵⁾
 - Able to tolerate $2\text{-}3\text{h/d}$ of skilled therapy $\geq 5\text{d/wk}$ and ≥ 1 therapy discipline
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **All:**

- Goal-directed therapy with rehabilitation potential ^(93, 94)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹⁵⁾
- Able to tolerate 2-3h/d of skilled therapy ≥ 5 d/wk and ≥ 2 therapy disciplines
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3 x/wk
 - Services, **All:**
 - Goal-directed therapy with rehabilitation potential ^(93, 94)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹⁵⁾
 - Able to tolerate ≥ 15 h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁹⁶⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁹⁷⁾
 - Ventilator dependent ≥ 6 h/d and weaning planned ^(98, 99)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽¹⁰⁰⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
061 ISCHEMIC STROKE, PRECEREBRAL OCCLUSION OR TRANSIENT ISCHEMIA WITH THROMBOLYTIC AGENT WITH MCC	n/a	5	CMS GMLOS
062 ISCHEMIC STROKE, PRECEREBRAL OCCLUSION OR TRANSIENT ISCHEMIA WITH THROMBOLYTIC AGENT WITH CC	n/a	3.2	CMS GMLOS
063 ISCHEMIC STROKE, PRECEREBRAL OCCLUSION OR TRANSIENT ISCHEMIA WITH THROMBOLYTIC AGENT WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
064 INTRACRANIAL HEMORRHAGE OR CEREBRAL INFARCTION WITH MCC	n/a	4.6	CMS GMLOS
065 INTRACRANIAL HEMORRHAGE OR CEREBRAL INFARCTION WITH CC OR TPA IN 24 HOURS	n/a	2.9	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
066 INTRACRANIAL HEMORRHAGE OR CEREBRAL INFARCTION WITHOUT CC/MCC	n/a	1.9	CMS GMLOS
STROKE	10%	4.7	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Transition plan: InterQual's® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan. There is a high risk of early and long-term hospital readmissions following a transient ischemic attack (TIA) and acute ischemic stroke. Following TIA and acute stroke, 17.4%(0.174) of patients will require readmission within the first month, and 42.5%(0.425) within the first year. Some factors associated with readmission include acute cerebrovascular disease, infection, occlusion or stenosis of pre-cerebral arteries, and acute renal failure (Yousufuddin et al., Journal of Stroke and Cerebrovascular Diseases 2019, 28: 167-74; Bambhroliya et al., JAMA network open 2018, 1: e181190-e; Vahidy et al., Stroke 2017, 48: 1386-8; Zhong et al., Neurological Sciences 2016: 1-8).

3:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a

decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

4:

Expected progress after 12-24h

- Neurological deficit stabilizing

Consider:

- Physical and occupational therapy evaluations
- Transition to PO medications
- Speech therapy/Nutrition consults if patient failed bedside swallowing screen
- Neurology consultation for progressive neurologic deficits
- Deep vein thrombosis (DVT) prophylaxis
- Continuous cardiac monitoring
- Neurological assessment at least 3x/24h

5:

Care Facilitation

Referral for psychiatric evaluation and/or treatment if co-occurring psychiatric symptoms are suspected or confirmed (e.g., depression, anxiety)

Knowledge deficit or non-adherence with condition management identified (**Home Care**)

Functional and/or cognitive impairment requiring additional therapy

Rehabilitation settings and DME should be considered

- At least 15h/wk of skilled therapies (**Acute Rehabilitation**)
- At least 2-3h for 5d/wk (**Subacute Rehabilitation**)
- At least 1-2h for 5d/wk, if the home environment is unsafe (**Skilled Nursing Facility**)
- Home with skilled therapy, if the home is safe and accessible (**Home Care**)

Prolonged ventilation and unable to wean

- Transfer to ventilator weaning program (**LTAC or SNF**)

Potential risk of readmission (**Home Care; Readmission reduction team**)

- TIA/stroke admission within the past month or first year post-TIA/stroke
- Knowledge deficit or non-adherence with condition management identified
- Co-morbid diabetes, hypertension, pneumonia, bleeding complications from anticoagulation, or nutritional deficits
- Fall risk/gait disturbance

End of life planning

- Consider palliative care for patients for end-stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be addressed throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients for end-stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meal programs, home or homemaking services, and transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or activity of daily living (ADL) impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on disease management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). 2019 exceptional surveillance of psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2019)
- In-home intervention to prevent medical relapse or readmission and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure, information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies, i.e., blood lead level testing and follow-up (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application and information should be given to patient
- Homeless shelter application and information should be given to patient

- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

6:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

7:

Patients presenting with minor non-disabling stroke are characterized by neurological symptoms that are resolving or have resolved. Examples of non-disabling symptoms may include dizziness, isolated sensory loss in limb or one side of the face, isolated diplopia, isolated facial droop, or isolated dysarthria (Mendelson and Prabhakaran, Jama 2021, 325: 1088-98; Edlow, Ann Emerg Med 2018, 71: 409-15; Wheatley, Emerg Med Clin North Am 2017, 35: 701-12). A neurological deficit resolved or resolving is defined as a patient's ability to (taking into account the patient's baseline functionality):

- Perform their activities of daily living (ADLs)
- Pass a bedside dysphagia screen
- Ambulate unassisted
- Score < 5 on the National Institute of Health Stroke Scale (NIHSS)

8:

Examples of non-disabling symptoms may include:

- Dizziness
- Isolated diplopia
- Isolated dysarthria
- Isolated facial droop
- Isolated sensory loss in limb or one side of face

9:

Patients presenting with minor non-disabling stroke are clinically evaluated for the source of neurologic compromise and for any risk of new or recurring neurological deficits. The following are appropriate diagnostic interventions performed after onset of symptoms: basic laboratory testing and urinalyses, cardiac monitoring for 12 hours, echocardiography, neuro-assessments, brain imaging with computerized tomography (CT) or magnetic resonance imaging (MRI) (preferably diffusion-weighted MRI (DW-MRI)), and vascular imaging with computed tomography angiography (CTA), magnetic resonance angiography (MRA), or carotid ultrasonography (Kleindorfer et al., Stroke 2021, 52: e364-e467; Powers et al., Stroke 2019, 50: e344-e418; Wheatley, Emerg Med Clin North Am 2017, 35: 701-12).

10:

Vascular imaging with computed tomography angiography (CTA), magnetic resonance angiography (MRA), or carotid ultrasonography is recommended early after diagnosis of minor non-disabling stroke to determine patients at increased risk of stroke who may benefit from carotid revascularization (Kleindorfer et al., Stroke 2021, 52: e364-

e467; Powers et al., Stroke 2019, 50: e344-e418).

11:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

12:

Suspected acute stroke refers to a patient whose clinical signs/symptoms or diagnostic imaging is indicative of an ischemic or hemorrhagic stroke.

13:

Instruction: Patients requiring thrombolytics or endovascular intervention should be managed at the Critical level of care.

14:

Expected treatments for patients experiencing acute ischemic stroke include (Kleindorfer et al., Stroke 2021, 52: e364-e467; Rothwell et al., Lancet 2016, 388: 365-75):

- Antiplatelet within 24 to 48 hours of symptom onset, dependent on use of thrombolytics
- Frequent assessment of neurologic status
- Deep vein thrombosis (DVT) prophylaxis
- Judicious blood pressure management
- Frequent turning of patients who are bedridden
- Aspiration risk evaluation and airway protection

15:

Instruction: These criteria refer to a persistent, new neurological deficit(s) from baseline that occurs as a result of the new admitting stroke or is an exacerbation of a neurological deficit from a prior stroke. Use of a standardized stroke scale, such as the National Institutes of Health Stroke Scale (NIHSS), can help measure the severity of a neurological deficit due to a stroke (Powers et al., Stroke 2019, 50: e344-e418).

16:

Abnormal movement pattern(s) and/or alteration(s) in muscle tone may include limb or trunk ataxia, flaccidity, hypotonicity, or apraxia.

17:

Alteration in speech may include dysarthria (slurred or garbled speech), dysphasia (difficulty in comprehension or expression), or aphasia (full loss of language).

18:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

19:

Paresis refers to a condition of muscle weakness with a reduction in voluntary movement while paralysis refers to a complete loss of movement. These criteria include, but are not limited to, paresis or paralysis of the face as well as difficulty swallowing (i.e., dysphagia).

Instruction: Patients experiencing general weakness or inability to move limbs from a non-neurologic cause are

excluded from these criteria.

20:

Sensory deficits due to stroke tend to be focal in nature and can include hemianesthesia, single limb anesthesia, facial hemianesthesia, hypoesthesia (i.e., decreased sensitivity), hyperesthesia (i.e., increased sensitivity), dysesthesia, or paresthesia (altered sensations). Numbness or loss of sensation, changes in sight or hearing are more often associated with ischemia versus other etiologies. The exception to this is with thalamic strokes which can present with pain. Thalamic involvement has been reported in up to 11%(0.11) of posterior infarcts (Gurley and Edlow, Acad Emerg Med 2019, 26: 1273-84).

21:

Visual impairment is common after a stroke and can include visual field deficits, sudden or temporary loss of vision (monocular visual loss or homonymous hemianopia), diplopia (double vision), oscillopsia, nystagmus, gaze preference, or visual neglect.

22:

Instruction: Application of this criteria point is intended for patients with acute vertigo or dizziness at rest due to a stroke. Other causes of vertigo or dizziness (e.g., benign paroxysmal positional vertigo (BPPV), Meniere's disease, vestibular neuritis, vestibular migraine) are excluded.

23:

Vertigo or dizziness are common symptoms with posterior circulation strokes and commonly associated with headache, gait disturbance and/or unilateral weakness depending on affected vascularity (Voetsch and Sehgal, Neurol Clin 2021, 39: 373-89; Gurley and Edlow, Acad Emerg Med 2019, 26: 1273-84; Tarnutzer et al., Neurology 2017, 88: 1468-77). Acute vestibular syndrome (AVS) is defined as the sudden onset of continuous dizziness/vertigo at rest, nausea/vomiting, nystagmus, head movement intolerance, and unsteady gait that persists longer than 24h. AVS can be caused by posterior circulation stroke or more commonly, by vestibular neuritis. It remains diagnostically crucial to identify and differentiate between the two processes where clinically appropriate based on findings and risk factors. It is important to note that a small percentage of patients with isolated cerebellar or brainstem stroke may or may not present with nystagmus in conjunction with dizziness or vertigo (Gurley and Edlow, Emerg Med Clin North Am 2021, 39: 181-201; Kattah, Stroke Vasc Neurol 2018, 3: 190-6).

24:

The primary goals of acute stroke imaging are to distinguish ischemic stroke from intracerebral hemorrhage, to select ischemic stroke patients for reperfusion therapies, and to exclude conditions that mimic acute cerebral ischemia. Both computed tomography (CT) and magnetic resonance imaging (MRI) may be used to evaluate acute neurologic changes; however, CT is more readily available. CT has the advantage of a shorter study time, and it can distinguish acute ischemic from hemorrhagic stroke. MRI is valuable in acute stroke patients with unusual presentations, stroke varieties, or when a stroke mimic is suspected but not clearly identified on CT. MRI with diffusion weighted imaging (DWI) has the highest sensitivity for detecting the presence, size, location, and extent of ischemia (Kleindorfer et al., Stroke 2021, 52: e364-e467; Powers et al., Stroke 2019, 50: e344-e418; Expert Panel on Neurologic et al., J Am Coll Radiol 2017, 14: S34-S61).

25:

A non-diagnostic brain scan can occur in a small percentage of patients with acute ischemic stroke. Although the use of diffusion-weighted imaging (DWI) has resulted in significant improvements in the detection of acute ischemic stroke versus head computed tomography (CT) due to its higher sensitivities, a significant but small number of patients with acute ischemic stroke may have a negative DWI. The three most common DWI-negative infarcts include posterior circulation ischemia, small strokes, particularly in the brainstem, and hyperacute ischemia (Edlow et al., Neurology 2017, 89: 256-62). Patients with posterior circulation ischemia may present with non-specific clinical characteristics including dizziness, vertigo, disequilibrium, visual symptoms, vomiting, confusion, etc. Patients who report dizziness with a posterior circulation ischemia have an even higher likelihood of a negative-DWI in the first 48 hours (Gurley and Edlow, Acad Emerg Med 2019, 26: 1273-84; Sommer et al., Stroke 2018, 49: 2728-32; Edlow et al., Neurology 2017, 89: 256-62; Tarnutzer et al., Neurology 2017, 88: 1468-77).

26:

Central nervous system (CNS) imaging within the first 24h of symptom onset is useful diagnostically and prognostically. Head computerized tomography (CT) and magnetic resonance imaging (MRI) can be useful in assessing for hemorrhage expansion (Greenberg et al., Stroke 2022, 53: e282-e361).

27:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

28:

Thrombolytic agents are medications that act to dissolve culprit clots. Indications for the use of thrombolytic agents include acute myocardial infarction, thrombotic stroke, massive pulmonary embolism, deep vein thrombosis, peripheral arterial thromboembolic occlusion, and vascular graft thrombosis. The most common complication of thrombolytic therapy is hemorrhage. Risk increases with the concomitant use of heparin or other antithrombotic agents. There is also a risk of embolism occurring from a dissolving clot.

29:

When intravenous (IV) thrombolytics are indicated, guidelines recommend administration as soon as possible as benefits are time dependent. Based on inclusion and exclusion criteria for selected patients, thrombolytics can be administered up to 3 hours from symptom onset or in some cases, from 3 hours up to 4.5 hours after the onset of symptoms (Powers et al., Stroke 2019, 50: e344-e418; Demaerschalk et al., Stroke 2016, 47: 581-641).

30:

Bleeding is a major complication associated with thrombolytic therapy. Intracranial hemorrhage (ICH) is the most common manifestation of bleeding and symptoms include change in level of consciousness, increased blood pressure, deterioration in motor exam, new onset of headache, or nausea and vomiting. Other common bleeding sites include retroperitoneal, genitourinary, and gastrointestinal. Minor bleeding such as oozing from gums, venipuncture sites, hematuria, and hemoptysis may occur. The patient should be monitored closely in the critical care setting for evidence of any signs of bleeding for the first 24 to 36 hours after intravenous (IV) thrombolytic administration.

31:

For patients with acute ischemic stroke who are eligible for intravenous recombinant tissue-type plasminogen activator (r-tPA), antiplatelet treatment should not be used as a substitute (Powers et al., Stroke 2019, 50: e344-e418).

32:

Major contraindications to thrombolysis include:

- Onset of symptoms, or last known to be at baseline, > 4.5 hours
- Head trauma, ischemic stroke, or myocardial infarction (excludes non-ST-segment elevation myocardial infarction) in previous 3 months
- Gastrointestinal malignancy or hemorrhage in previous 3 weeks
- Urinary tract hemorrhage in previous 3 weeks
- Major surgery in previous 2 weeks
- Intracranial or intraspinal surgery in previous 3 months
- Arterial puncture performed on a non-compressible site in past week
- Acute intracranial or subarachnoid hemorrhage
- History of intracranial hemorrhage
- Systolic blood pressure > 185 mmHg or diastolic blood pressure > 110 mmHg
- Active bleeding or acute trauma
- Treatment with low-molecular-weight heparin (excludes prophylactic dose) within last 24 hours
- INR $\geq$ 1.7

- PTT abnormal
- Platelet count < 100,000 mm³
- Blood glucose < 50 mg/dL (2.7 mmol/L)
- Seizure with postictal residual neurological impairment
- Multilobar infarction
- Infective endocarditis
- Aortic arch dissection (Powers et al., Stroke 2018, 49: e46-e110)

33:

Patients receiving intravenous recombinant tissue-type plasminogen activator (r-tPA) therapy within 4.5 hours may additionally be eligible for endovascular therapy with a stent retriever within 6 to 24 hours of symptom onset. For selected patients who are no longer candidates for r-tPA but present within 6 to 24 hours of symptom onset, endovascular therapy with a stent retriever for an anterior circulation large vessel clot may be reasonable. Endovascular therapy with a stent retriever is recommended over intra-arterial fibrinolysis as a first line therapy (Lin et al., JAMA Neurol 2019, 76: 850-4; Powers et al., Stroke 2019, 50: e344-e418).

34:

In patients who are eligible for endovascular treatment, imaging studies of the following vasculature may provide useful information on patient eligibility and procedural planning (Powers et al., Stroke 2018, 49: e46-e110):

- Imaging of the extracranial carotid and vertebral arteries
- Imaging of intracranial circulation

35:

Adequate perfusion supporting organ function should be maintained by correcting hypovolemia and hypotension in the patient with stroke (Powers et al., Stroke 2019, 50: e344-e418).

36:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous pressure, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

37:

Blood pressure (BP) management for stroke patients is dependent upon multiple factors and requires clinical decision-making by the physician to individualize treatment. Stroke guidelines recommend lowering systolic blood pressure (SBP) to less than 185 mmHg and diastolic blood pressure (DBP) to less than 110 mmHg before administration of intravenous (IV) alteplase or performing thrombectomy. Close BP monitoring after IV thrombolysis is recommended with treatment aimed at lowering BP to less than 180/105 mmHg for the first 24h (Powers et al., Stroke 2019, 50: e344-e418). There have been limited clinical trials for hypertension management in patients with BP greater than or equal to 220/120 mmHg who do not receive IV or mechanical thrombolysis. Current guidelines recommend lowering BP by 15%(0.15) in the first 24 hours after onset of stroke in these patients (Warner et al., Stroke 2019, 50: 3331-2). The SBP target is less than 140 mmHg with a maintenance SBP between 130 mmHg and 150 mmHg in patients with hemorrhagic stroke of mild to moderate severity (Glasgow Coma Scale score greater than or equal to 5) who present with a SBP between 150 mmHg and 220 mmHg. In patients with hemorrhagic stroke of mild to moderate severity presenting with SBP greater than 150 mmHg, lowering SBP less than 130 mmHg is potentially harmful (Greenberg et al., Stroke 2022, 53: e282-e361).

38:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities,

clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., *PulmCCM* 2018; Nardi et al., *F1000Res* 2017, 6: 290; Osadnik et al., *Cochrane Database Syst Rev* 2017, 7: CD004104; Allison and Winters, *Emerg Med Clin North Am* 2016, 34: 51-62; Gregoretti et al., *Crit Care Clin* 2015, 31: 435-57).

39:

Studies combining clopidogrel and aspirin have been shown to decrease the risk of ischemic events compared to aspirin alone in patients who have experienced minor strokes or high risk transient ischemic attack. This combination therapy does, however carry an increase in the risk of major and minor hemorrhage (SC et al., *N Engl J Med* 2018, 379: 215-25).

40:

Guidelines strongly recommend oral anticoagulants for patients with atrial fibrillation (AF). Non-vitamin K antagonist oral anticoagulants (NOACs), such as dabigatran, rivaroxaban, apixaban, and edoxaban, are recommended over warfarin in NOAC eligible patients with AF (e.g., those without contraindications to specific medications and without moderate to severe mitral stenosis or a mechanical heart valve). Warfarin anticoagulation is appropriate for patients with valvular AF. Anticoagulants are contraindicated for hemorrhagic stroke (Kleindorfer et al., *Stroke* 2021, 52: e364-e467; Beaser and Cifu, *Jama* 2019, 321: 1100-1; Kernan et al., *Stroke* 2014, 45: 2160-236).

41:

Following an ischemic stroke, antiplatelet administration is recommended within 24 to 48 hours after symptom onset. For patients treated with intravenous recombinant tissue-type plasminogen activator (r-tPA), aspirin is generally delayed for 24 hours (Uchino, *Annals of internal medicine* 2019, 170: JC38; Powers et al., *Stroke* 2018, 49: e46-e110). Antiplatelet agents are contraindicated for hemorrhagic stroke.

42:

In patients with stroke, oral anticoagulants are indicated in those patients with embolic disease resulting from atrial fibrillation (Kleindorfer et al., *Stroke* 2021, 52: e364-e467; Uchino, *Annals of internal medicine* 2019, 170: JC38; Beaser and Cifu, *Jama* 2019, 321: 1100-1). Anticoagulants are contraindicated for hemorrhagic stroke.

43:

All anti-platelet agents increase the risk of hemorrhage and are contraindicated in patients who have thrombocytopenia, gastric ulcer, severe hypertension, or an active or recent hemorrhagic event (e.g., intracranial bleeding, trauma, or major surgery within the past 30 days).

44:

The exact blood pressure (BP) at which the risk of hemorrhage after thrombolysis increases is unknown. Guidelines recommend patients who have elevated BP and are otherwise eligible for thrombolytics should have their BP carefully lowered so that their systolic blood pressure (SBP) is less than 185 mmHg and their diastolic blood pressure is less than 110 mmHg prior to the initiation of intravenous fibrinolytic therapy. A target SBP of less than 180 mmHg and DBP of less than 105 mmHg should be maintained for 24 hours following reperfusion therapy (Powers et al., *Stroke* 2019, 50: e344-e418). The safety of more rapid BP reduction following thrombolysis is presently being investigated (Anderson et al., *The Lancet* 2019, 393: 877-88).

45:

Corticosteroids are contraindicated for post-stroke cerebral edema (Powers et al., *Stroke* 2018, 49: e46-e110).

46:

Hyperosmolar therapy creates an osmotic gradient, facilitating the removal of water across the blood brain barrier, resulting in a decreased brain volume and a decrease in intracranial pressure. While hypertonic saline and mannitol are the 2 most used hyperosmolar agents, there is insufficient evidence to recommend one over the other for improving neurological outcomes in patients with acute ischemic stroke (Cook et al., Neurocrit Care 2020, 32: 647-66). Doses and strengths are generally determined by institutional practice patterns. Common side effects associated with osmotic therapy include dehydration, hypotension, renal failure, and acute respiratory distress.

47:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services, Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

48:**Expected progress after 24-48h**

- Neurological deficit stabilizing

Consider:

- Physical and occupational therapy evaluations
- Transition to PO medications
- Speech therapy/Nutrition consults if patient failed bedside swallowing screen
- Fall risk prevention
- Neurology consultation for progressive neurologic deficits
- Physical activity progression
- Infectious Disease consultation for persistent fever

49:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

50:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

51:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

52:

Secondary prevention for stroke or transient ischemic attack (TIA) includes management of hypertension. For patients experiencing a stroke or TIA that were not previously receiving antihypertensive medication therapy, initiation is suggested for systolic blood pressure (SBP) greater than or equal to 140 mmHg, or a diastolic blood pressure (DBP) greater than or equal to 90mmHg. Antihypertensive therapy should be resumed for those patients previously receiving drug therapy in order to prevent recurrent stroke and other vascular events.

The following lifestyle alterations are helpful in the reduction of blood pressure:

- Salt restriction

- Weight loss
- Increase fruit and vegetables in the diet
- Decrease fat intake
- Limit alcohol intake
- Regular physical activity
- Smoking cessation

Goals for a target blood pressure should be individualized. However, managing the blood pressure to a SBP less than 140 mmHg and DBP less than 90 mmHg during the recovery period is recommended (Kleindorfer et al., Stroke 2021, 52: e364-e467; Powers et al., Stroke 2019, 50: e344-e418).

53:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

54:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

55:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

56:

Reference: (Powers et al., Stroke 2019, 50: e344-e418)

57:

Patients who develop bleeding while receiving anticoagulation should be evaluated for an underlying source of bleeding. Evaluation may include upper and lower gastrointestinal (GI) studies to rule out underlying GI pathology or an imaging study to rule out intracranial bleeding (Janz, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 9 edn. 2018:1463-79.e3).

58:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

59:

Orthostatic hypotension (Juraschek et al., JAMA Intern Med 2017, 177: 1316-23).

60:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to

ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory status. The patient will require assistance with one or more limitations.

61:

Instruction: These criteria refer to a functional impairment that occurs as a result of the admitting new stroke or is an exacerbation of a functional impairment from a prior stroke.

62:

Rehabilitation may include, but is not limited to, physical therapy, occupational therapy, or speech therapy.

63:

Instruction: This criterion includes aspiration pneumonitis. Aspiration is marked by the abnormal movement of oropharyngeal or gastric contents into the lower respiratory tract. Aspiration pneumonitis occurs when acidic gastric contents cause inflammation of the lung parenchyma. In the early stages, aspiration pneumonitis is not considered an infectious process as gastric acid prevents microbial growth. Anti-infective therapy should be considered if symptoms (e.g., shortness of breath, wheezing, hypoxemia) fail to resolve within 48 hours (Neill and Dean, Curr Opin Infect Dis 2019, 32: 152-7).

64:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

65:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

66:

Expected treatments for patients with bacterial pneumonia include anti-infective and oximetry or blood gas monitoring (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Cohoon et al., Am J Med 2018, 131: 307-16 e2; Kaysin and Viera, Am Fam Physician 2016, 94: 698-706; Jain et al., N Engl J Med 2015:[e-pub]).

67:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

68:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

69:

Routine supplemental oxygen is not automatically required for acute stroke patients. Guidelines recommend that oxygen only be administered to hypoxic stroke patients with an oxygen saturation less than 94%(0.94) (Powers et al., Stroke 2019, 50: e344-e418).

70:

Refractory seizures, also known as uncontrolled, intractable, or drug-resistant seizures, are resistant to treatment and are poorly controlled with medication. Recurrent seizures are seizures that occur more than once and may occur without warning (Billington et al., J Clin Med 2016, 5:).

71:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services, Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

72:**Expected progress 48-72h**

- Neurological deficit stable or improving
- Nutritional route established
- IV medications converted to PO
- Vital signs within acceptable limits
- Increasing activity
- Post-acute therapy arranged based on functional needs/impairment
- Discharge planning

If not, consider:

- Repeat brain imaging (magnetic resonance imaging (MRI) or computed tomography (CT)) and neurology consultation for progressive neurologic deficits
- Need for feeding tube placement

73:

Hypertension is defined as a systolic blood pressure (SBP) of greater than 130 mmHg or a diastolic blood pressure (DBP) of 80 mmHg or greater. Recommendations for initial management may include a conservative approach with diet and lifestyle modification with or without medications. The goal of antihypertensive therapy is to reduce morbidity and mortality. In non-urgent situations, blood pressure (BP) control is achieved in the outpatient setting. Goals for target BP level should be individualized, but a SBP less than 130 mmHg and a DBP less than 80 mmHg are desirable. Populations at high risk include patients with the following conditions (Whelton et al., Hypertension 2018, 71: 1269-324):

- Clinical cardiovascular disease or 10-year atherosclerotic cardiovascular disease (ASCVD) risk 10%(0.1)
- Heart failure
- Stable ischemic heart disease
- Chronic kidney disease
- Chronic kidney disease after renal transplantation

74:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

75:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

76:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

77:

Reference: (Powers et al., Stroke 2019, 50: e344-e418)

78:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

79:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

80:

The following discharge instructions may be individualized as appropriate (Gomez et al., F1000Res 2017, 6: 1893; Winstein et al., Stroke 2016, 47: e98-e169; Kernan et al., Stroke 2014, 45: 2160-236):

- Call emergency medical services immediately if there is any sudden weakness or numbness of the face, arm, or leg; blurry or decreased vision; inability to speak or understand; dizziness, loss of balance or falling; severe headache
- Take all medications as prescribed
- Eat a well-balanced diet low in cholesterol and saturated fat; drink 6-8 glasses of water or juice every day

to maintain good hydration and avoid constipation

- Have blood pressure checked regularly as uncontrolled high blood pressure is a risk factor for stroke
- Diabetics should maintain tight control of their blood glucose since elevated blood glucose contribute to stroke
- Consume alcohol in moderation
- Smokers should consider participation in a smoking cessation program

81:

Patients may require outpatient rehabilitation or home care services depending on their functional limitations and impairments. Rehabilitation services may include physical therapy (PT), occupational therapy (OT), speech therapy (ST), cognitive training or vocational rehabilitation, splinting, bracing, or durable medical equipment to enhance quality of life in the face of residual disability.

Other areas that require evaluation and support include:

- Evaluation by the caregiver for signs which may include: feelings of persistent sadness, weight fluctuations (due to overeating or lack of appetite), fatigue, irritability, persistent physical symptoms that do not respond to treatment
- Evaluation by the caregiver for evidence of clinical deterioration. Emphasis must be placed on the importance of early entry into the hospital system for prompt treatment and to allow the potential for resolution of symptoms
- Lifestyle changes to prevent recurrent ischemic stroke which include regular medical checkups, control of blood pressure, achieving and maintaining a healthy weight, regular exercise, maintaining control of diabetes, and reducing stress
- Achieving maximum independence through participation in recommended therapy programs

82:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

83:

Based on the results from the Stroke Prevention by Aggressive Reduction in Cholesterol Levels (SPARCL) trial, high-intensity statin therapy, when appropriate, is recommended for all patients with ischemic stroke and transient ischemic attack to reduce the risk of stroke and cardiovascular events (Grundy et al., Circulation 2019, 139: e1082-e143).

84:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

85:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

86:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

87:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

88:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

89:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

90:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

91:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

92:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

93:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening

- Wheelchair mobility
- Ambulation

94:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

95:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

96:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

97:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

98:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

99:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia

- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

100:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784).

Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.